Section Report 3 — Independent Amplicon Analysis

BIOL 4810

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Section Report 3 — Independent Amplicon Analysis

Goal: Use your **own sequencing dataset** to complete the full workflow in Galaxy: process amplicon reads with **DADA2**, then visualize your results in **phyloseq** using **alpha diversity**, **beta diversity**, and a **taxonomic bar plot**.
Submission: Fill in answers, upload screenshots where prompted, then click **Prepare submission preview** and **Download as PDF (Print)**.

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0) Your dataset

0A. Sample set / project name:

Bird dataset

0B. Sequencing type:

paired-end Illumina

0C. Metadata grouping variable you analyzed:

population

Workflow overview

For this section report, you will complete:

```
plotQualityProfile → filterAndTrim → learnErrors → dada → mergePairs →
makeSequenceTable → removeBimeraDenovo → assignTaxonomy → phyloseq alpha
diversity → phyloseq beta diversity (NMDS using Bray-Curtis) → phyloseq
taxonomic bar plot
```

Reminder: This section report asks you to apply the entire workflow to your own data and explain the biological meaning of each major output.

Part 1 — Quality profiles

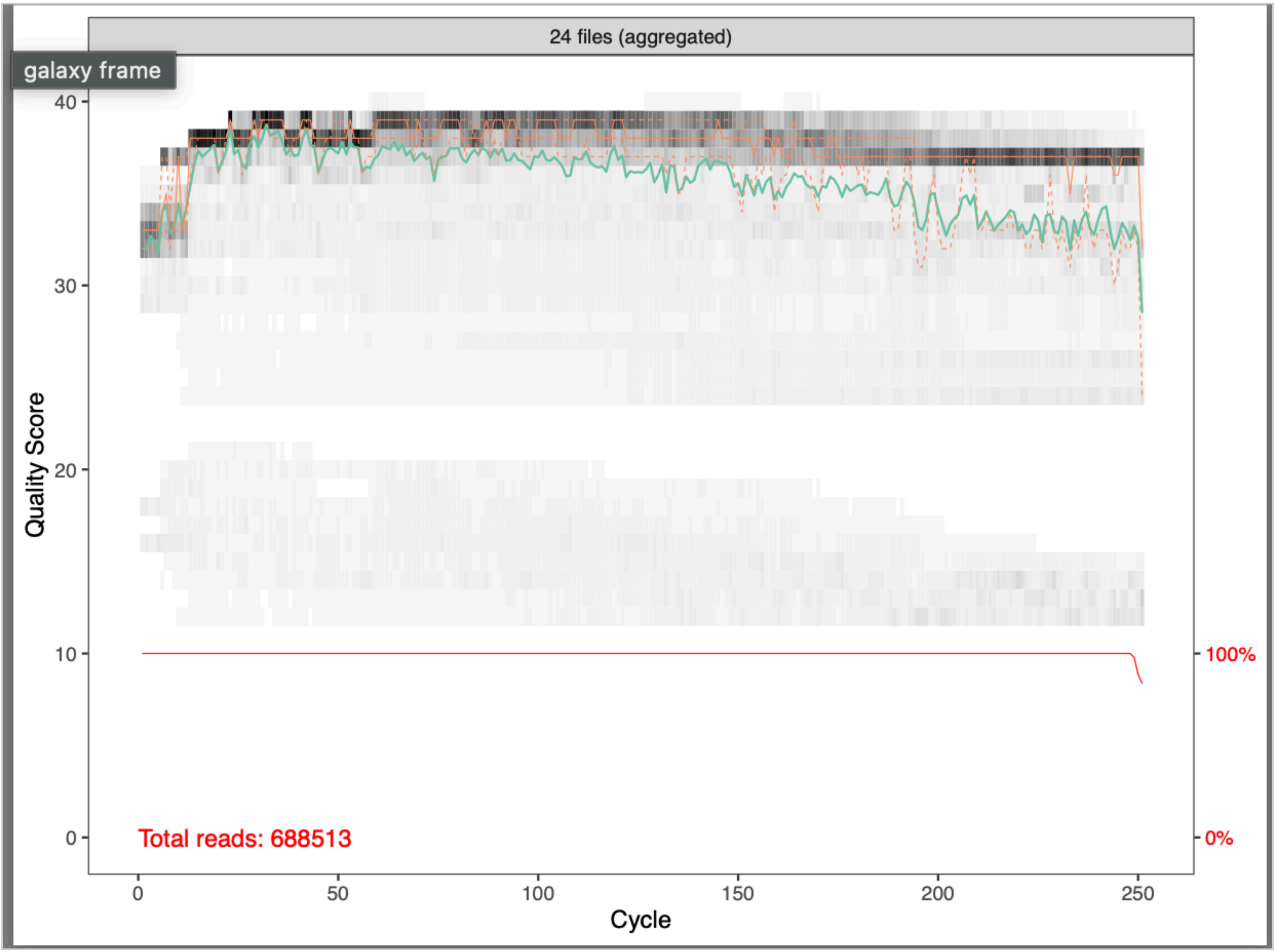
1A) Forward read quality profile

What is the purpose of this step?

The quality profile tool aids in visualizing and understanding the distribution of respective quality scores.

Upload screenshot: forward plotQualityProfile PDF

Choose File forwardreads



Where does quality begin to noticeably decline in the forward reads?

210

Chosen forward truncation length:

220

1B) Reverse read quality profile

What is usually different about reverse-read quality compared with forward-read quality?

Reverse reads are typically lower quality than forward reads because by the time the first read has been sequenced and the second can start, the signal has degraded and accuracy declines.

Upload screenshot: reverse plotQualityProfile PDF

Choose File reversereads

Where does quality begin to noticeably decline in the reverse reads?

170

Chosen reverse truncation length:

200

1C) Truncation decision

Explain why you chose these truncation values.

I chose truncation values a margin away from those values whereby quality began to drop in order to keep as much overlap as possible for merging. Willing to tolerate maybe a quality score above 25.

Rule of thumb: trim where quality begins to drop, but still keep enough overlap for forward and reverse reads to merge.

Part 2 — filterAndTrim

What filterAndTrim does:
This step removes low-quality reads and trims poor-quality bases from the ends of reads before denoising.

Why is this step important?

Trimming from the ends and filtering enhances the accuracy of downstream processing by helping to eliminate sequencing errors.

Parameters used:

trimming parameters “truncate read length” set to 220; yes to separate filters and trimmers for reverse reads; Trim start of each read set to 10; Truncate read length set to 200;

Upload screenshot: filterAndTrim output

Choose File Screenshot… 23.28.26

What could happen if you trim too aggressively?

Poor merging and loss of usable data can result from overtrimming, reducing coverage/depth of sequence(s).

What could happen if you do not trim enough?

Low quality sequences can increase error rates.

IMPORTANT — Split Paired Reads (Unzip Collection)

After `filterAndTrim`, your output is a **paired collection**.
Before continuing, you must split this into **forward and reverse collections**.

What to do in Galaxy

1. In the Tools panel, search: **unzip collection**
2. Input: your `filterAndTrim` paired output
3. Run

What you will get

- One collection = **forward reads (R1)**
- One collection = **reverse reads (R2)**

Why this matters

`learnErrors` and `dada` must be run **separately** on forward and reverse reads.
If you skip this, `mergePairs` will fail.

Part 3 — learnErrors

What learnErrors does:
This step models the sequencing error rates in your dataset so DADA2 can distinguish real sequence variants from sequencing mistakes.

Why is this step important?

The learnErrors step is important because it determines how and how often errors occur.

Upload screenshot: learnErrors PDF

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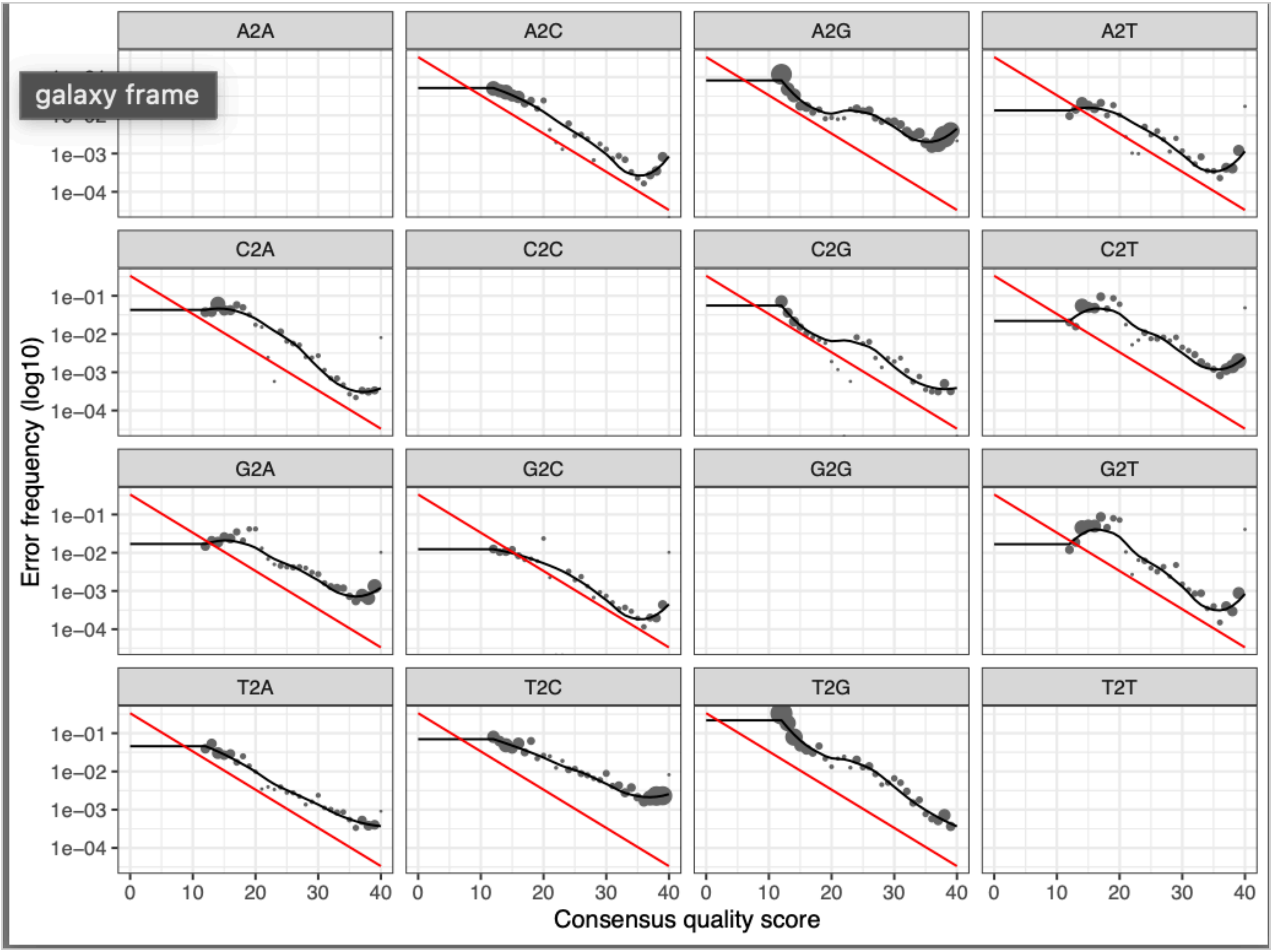


Does the fitted model generally follow the observed error pattern?

The fitted model mostly follows the observed error pattern.

Why is learning error rates important before denoising?

Learning error rates is important to perform before denoising because it allows dada2 to distinguish between true sequence variation and sequencing errors.

Part 4 — dada (denoising)

What dada does:
This is the denoising step. DADA2 uses the learned error model to correct likely sequencing errors and infer exact amplicon sequence variants (ASVs).

What is denoising doing?

When provided filtererd reads and error reads, it uses the learned error model to correct errors and reconstruct true sequences (amplicon sequence variants).

How is ASV inference different from OTU clustering?

OTU clustering groups of sequences into taxa based on similarity threshold without modeling errors, whereas ASV Inference corrects sequencing errors at the nucleotide level. ASV provides more precision in VL's opinion.

Part 5 — mergePairs

What mergePairs does:
This step merges the forward and reverse reads for each amplicon so you can reconstruct the full sequence from overlapping read pairs.

What is being merged, and why?

Forward and reverse reads are being merged to assess confidence of sequence accuracy.

Why do forward and reverse reads need to overlap?

Overlap between forward and reverse reads confirms that they match and indicates accuracy.

If reads fail to merge, what is one likely reason?

Reads may fail to merge if there isn't enough overlap.

Part 6 — makeSequenceTable

What makeSequenceTable does:
This step creates the ASV table. Each row is a sample, each column is an ASV, and each cell contains the number of times that ASV was observed in that sample.

What does this table represent biologically?

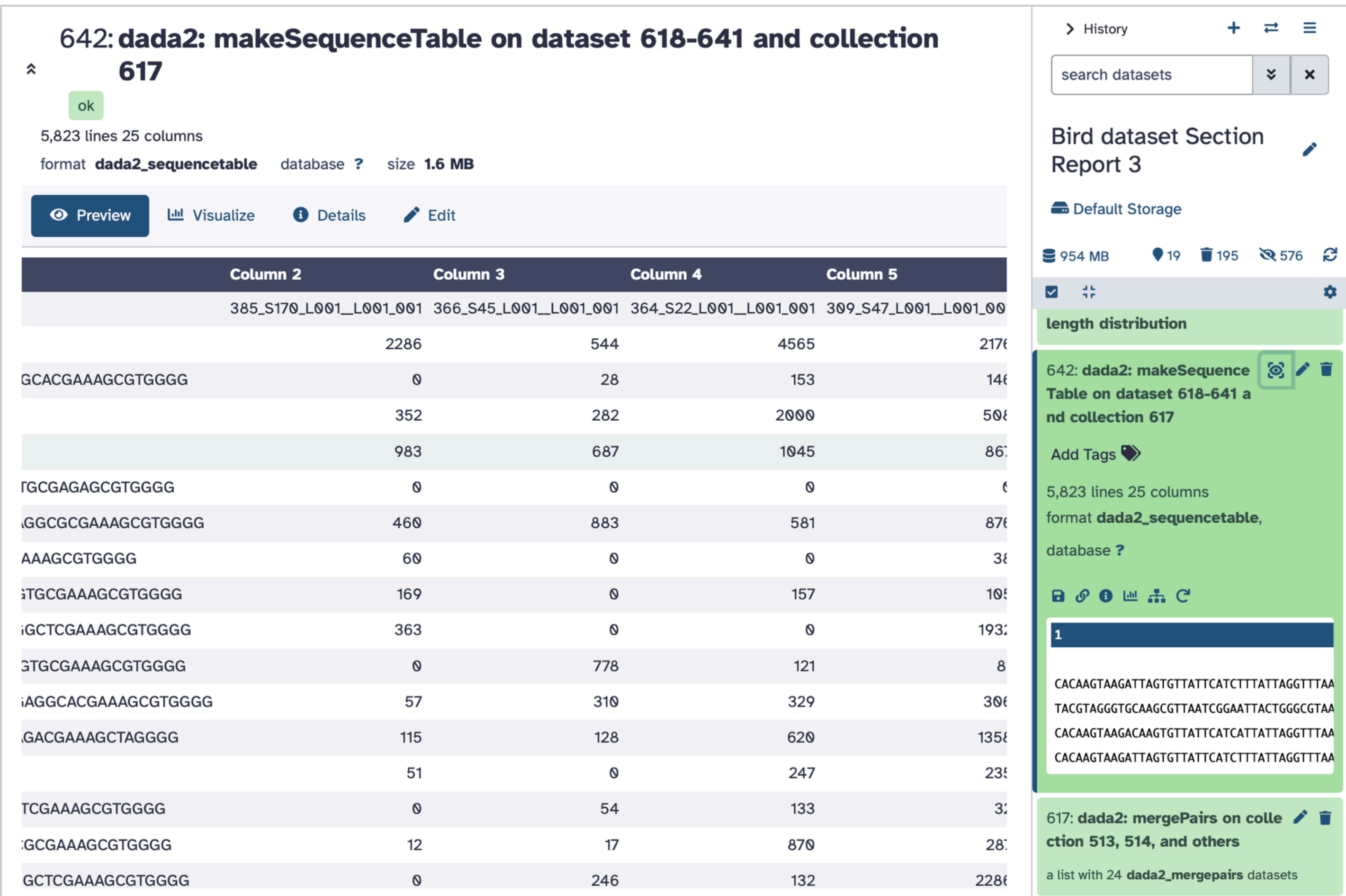
The ASV table represents diversity with precise comparison of sampled sequences. there are about 1600 ASVs.

What are the rows, columns, and cell values in the sequence table?

Columns are ASVs, samples, and frequency of ASV appearance in each sample.

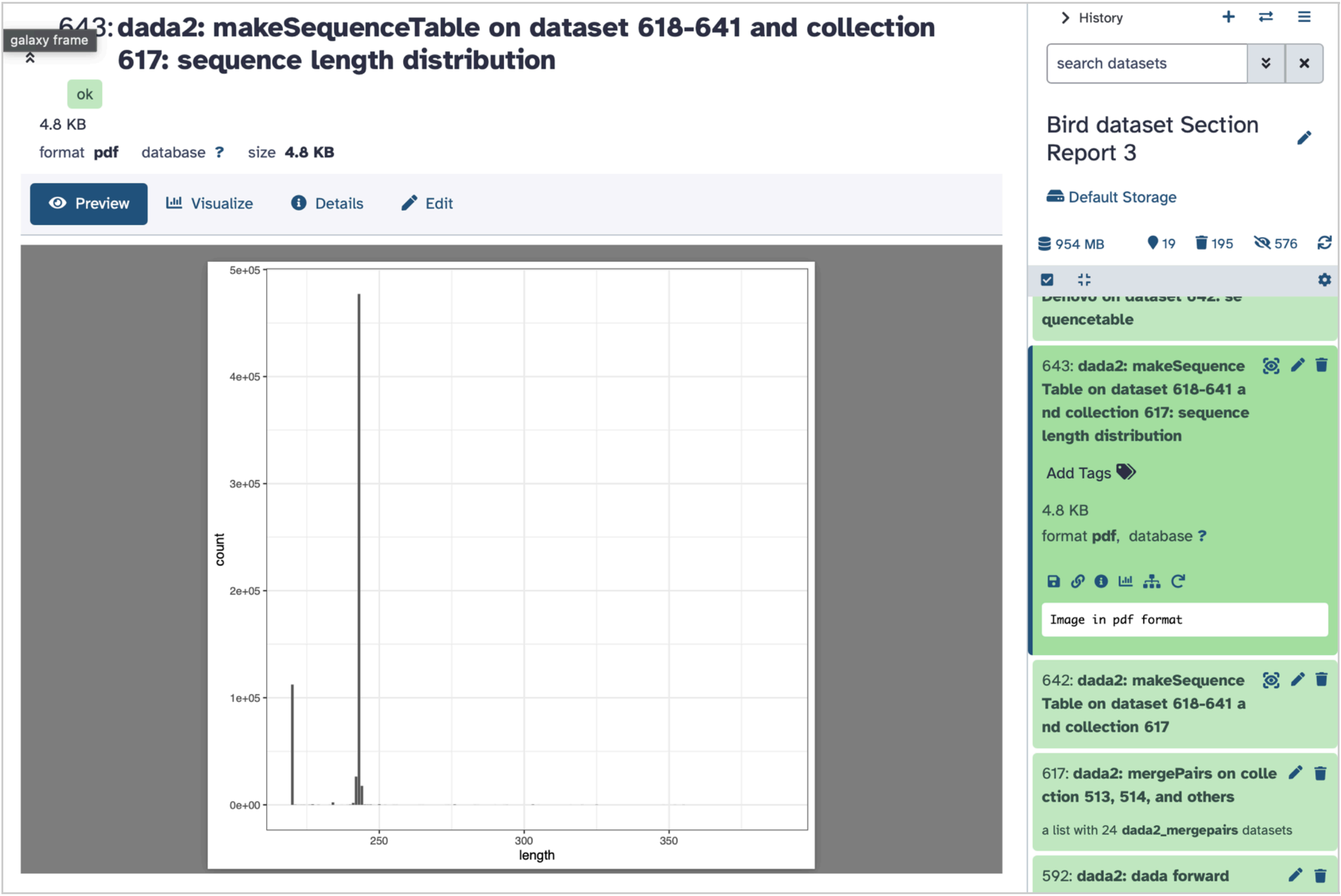
Upload screenshot: sequence table output

Choose File Screenshot… 23.56.27



Upload screenshot: sequence length distribution

Choose File Screenshot… 23.57.12



What does the sequence length distribution show?

Narrow distribution centered mostly around a single peak suggests that the target region was amplified fairly consistently between samples.

Part 7 — removeBimeraDenovo

What removeBimeraDenovo does:

This step removes chimeras, which are artificial sequences created during PCR when fragments from different templates are incorrectly joined together.

What is a chimera, and why should it be removed?

Chimeras are sequences artificially generated during PCR and because they don't actually exist in the original sample the presence of chimeras compromises sequence accuracy and downstream processing.

Why are chimeras a problem for interpretation?

The presence of chimeras can lead to incorrect taxonomic assignment.

How many chimeric reads were removed? How do you know?

40 chimeric reads were removed because there were 5,823 lines in the dada2makeSequenceTable file and 5,783 lines in the dada2removeBimeraDenovo file.

Part 8 — assignTaxonomy

What assignTaxonomy does:
This step compares your ASVs to a reference database and assigns taxonomy, such as phylum, class, order, family, genus, or sometimes species.

What does taxonomy assignment add to your ASV table?

Taxonomy assignment to ASV visually reflects the species and community composition from samples, as well as their roles.

Why might some ASVs not be assigned all the way to species?

Some ASVs may not be assigned all the way to species due to sequence resolution, reference database limitations, or primarily sequenced regions being too short.

Part 9 — Alpha diversity

9A) Alpha diversity

In your own words, what is alpha diversity?

Alpha diversity measures and describes how diverse a particular community is.

9B) Observed richness

What does Observed measure?

Observed species richness simply measures the total number of species.

9C) Shannon diversity

What does Shannon measure?

Shannon (alpha) diversity accounts for richness and evenness by comparing species abundance.

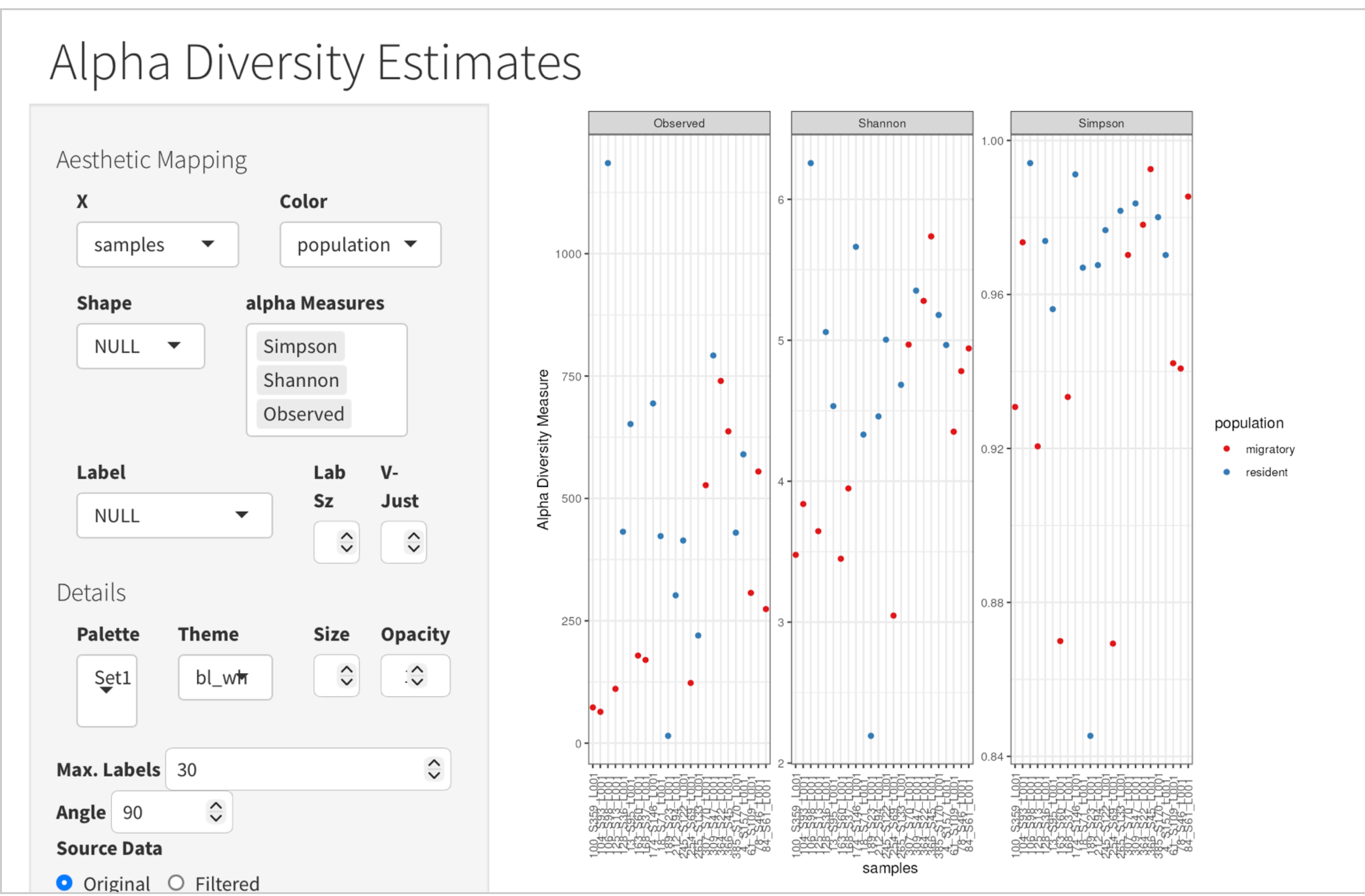
9D) Simpson diversity

What does Simpson measure?

Simpson (alpha) diversity accounts for richness and evenness with less sensitivity to rare species, essentially indicating the probability of picking the same species twice.

Upload screenshot: alpha diversity plot grouped by population

Choose File alphadiv

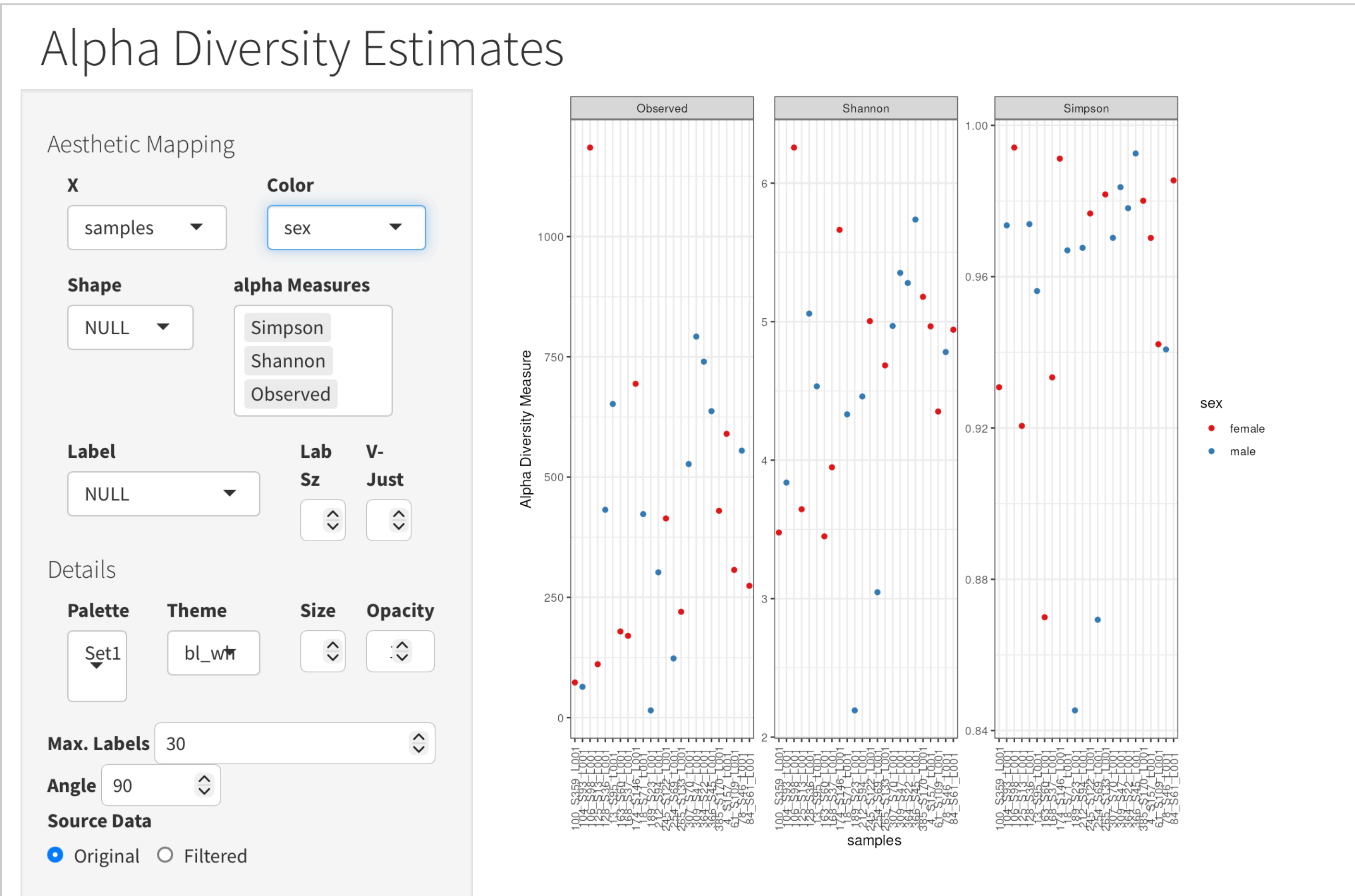


Describe differences in alpha diversity by population.

Generally, residents had a higher alpha diversity than migratory birds.

Upload screenshot: alpha diversity plot grouped by sex

Choose File alphadivsex

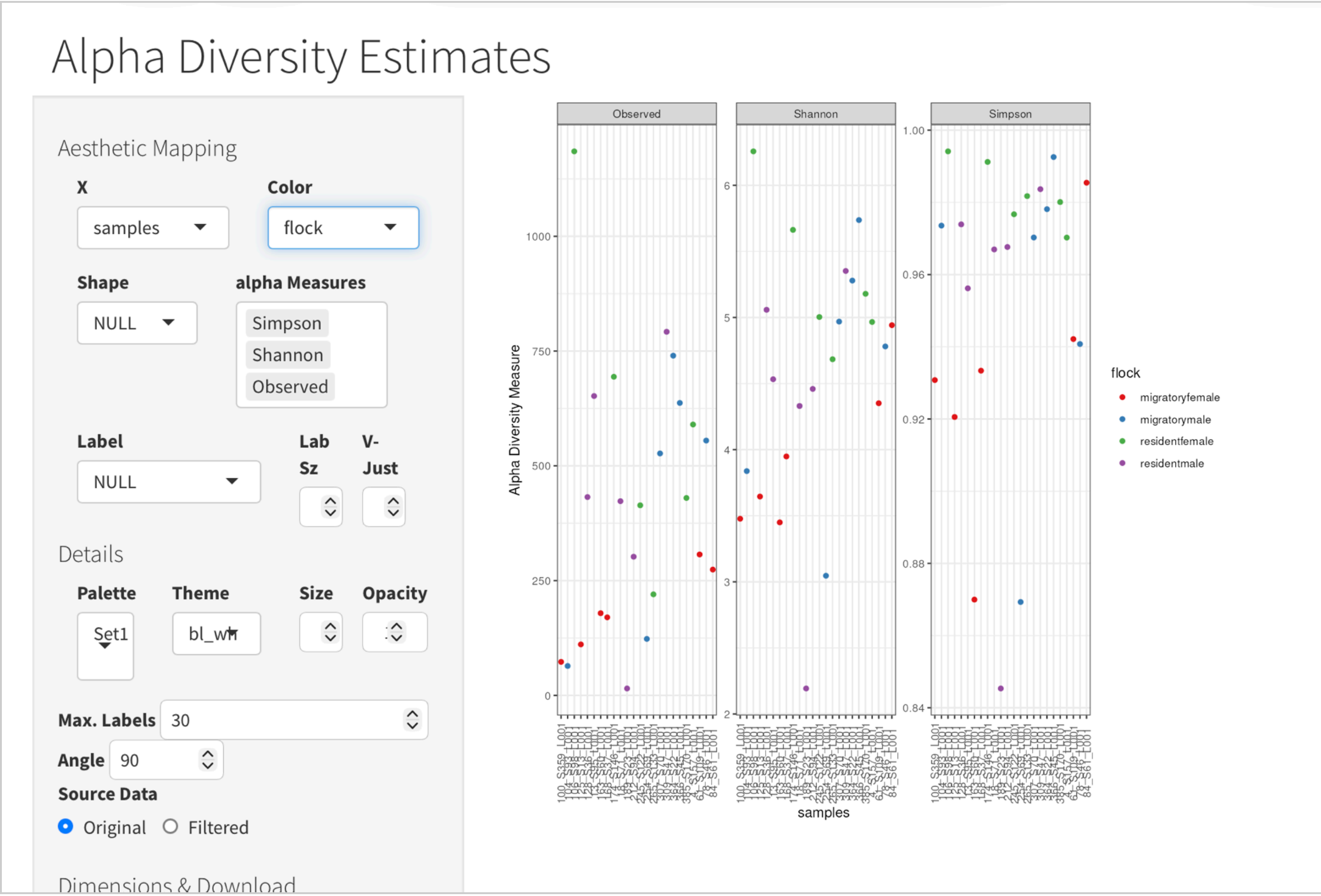


Describe differences in alpha diversity by sex.

Alpha diversity was lower for female birds than male, on average, with there being numerous exceptions.

Upload screenshot: alpha diversity plot grouped by flock

Choose File alphadivflock



Describe differences in alpha diversity by flock.

If there is any pattern to alpha diversity values here, it shows that flocks very roughly group together.

Part 10 — Beta diversity (NMDS with Bray-Curtis)

What beta diversity means:

Beta diversity describes differences in community composition **between samples**.

In your own words, what is beta diversity?

Beta diversity is a comparison of one community's composition to that of another.

10A) Bray-Curtis dissimilarity

What does Bray-Curtis measure?

Individuals (represented by datapoints) with similar microbiomes (respective datapoints will be closer) or dissimilar (respective datapoints will be farther)

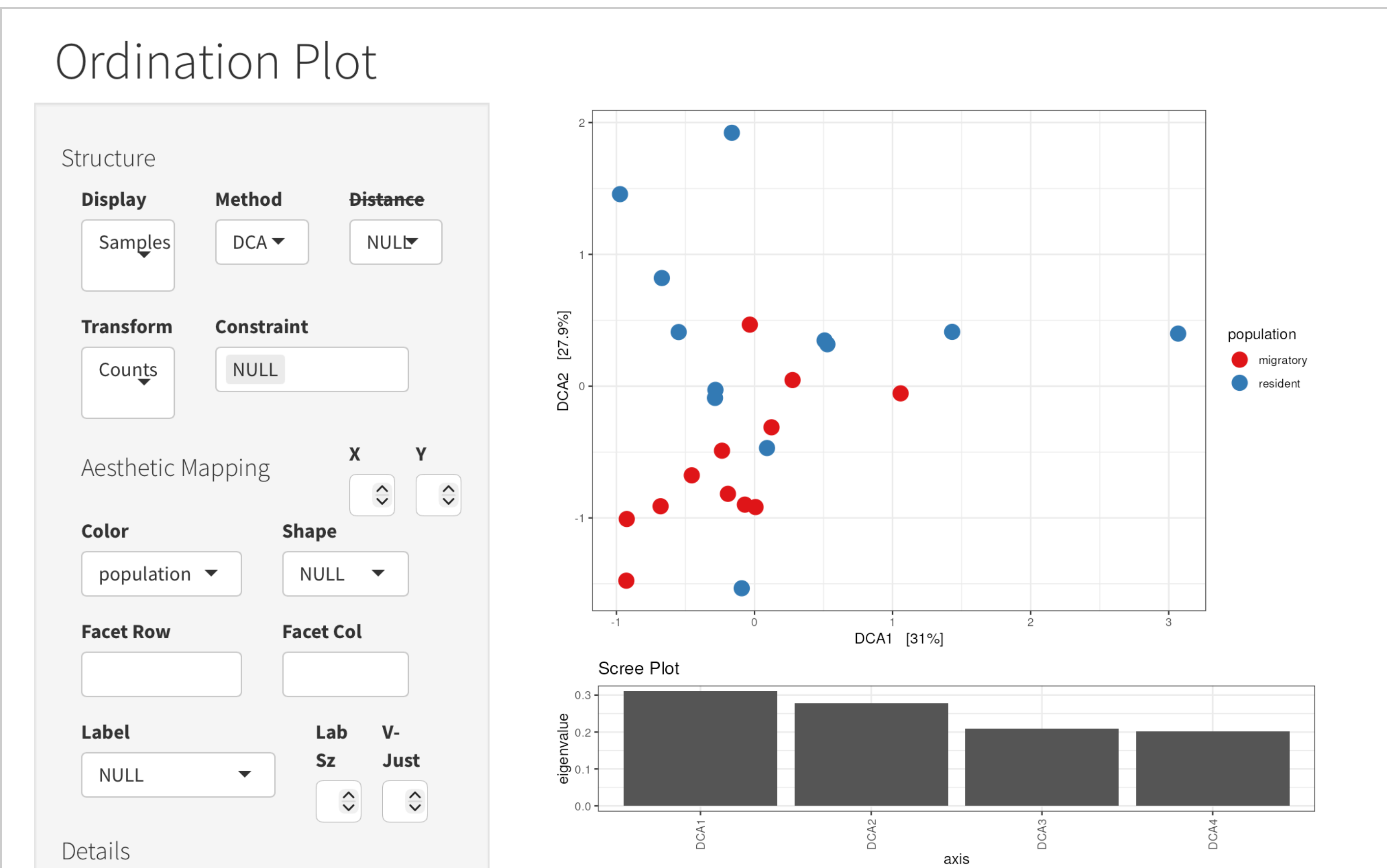
10B) NMDS

What does NMDS show?

The NMDS plot shows approximate similarity by sample source in this case, and more generally represents dissimilarity by datapoint proximity.

Upload screenshot: NMDS Bray-Curtis plot colored by population

Choose File nmdspop

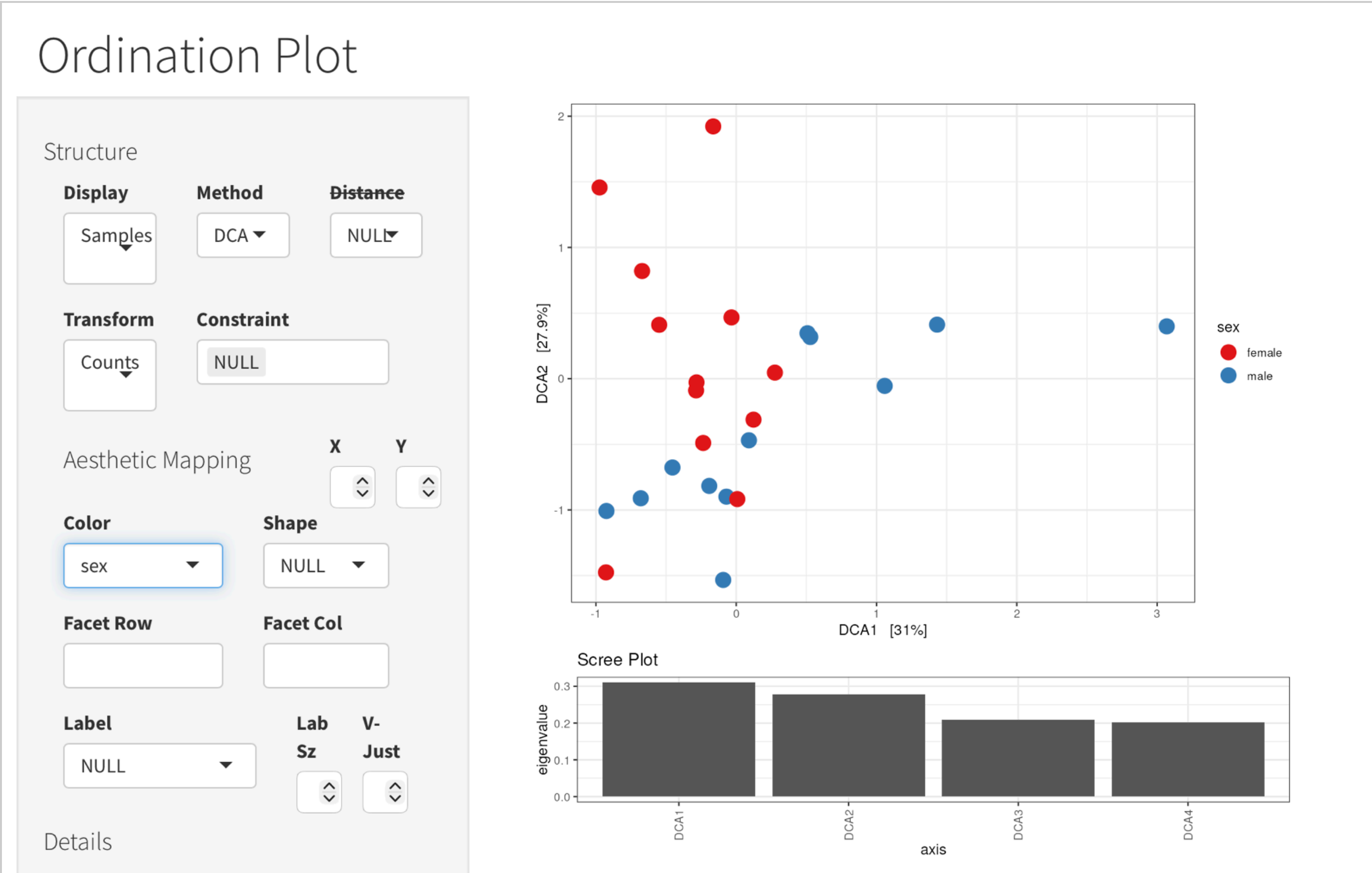


Describe beta diversity patterns by population.

The plot shows that there is grouping by beta diversity values around population status.

Upload screenshot: NMDS Bray-Curtis plot colored by sex

Choose File nmdssex



Describe beta diversity patterns by sex.

Grouping is less obvious here, but still visible on the basis of sex.

Upload screenshot: NMDS Bray-Curtis plot colored by flock

Choose File nmdsflock

Describe beta diversity patterns by flock.

Flocks roughly group together around their beta diversity values, with more obvious grouping noticeable from migratory flocks.

Part 11 — Taxonomic bar plot

What a taxonomic bar plot shows:

A bar plot summarizes the composition of samples by showing the relative abundance of taxa at a selected taxonomic level.

What taxonomic level did you use?

phylum

How did you facet the plot?

faceted rows by population

Why was this taxonomic level useful?

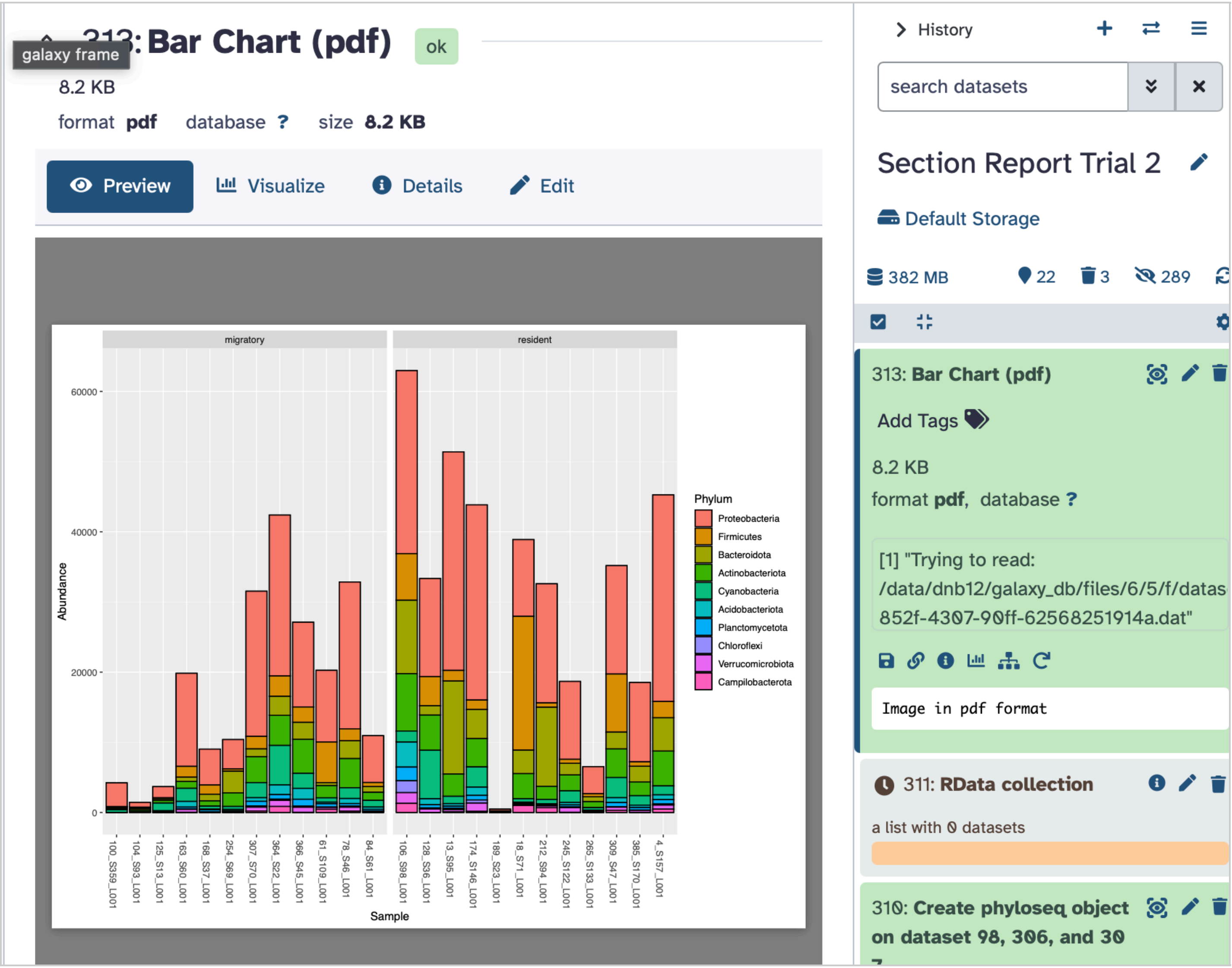
Using phylum as the taxonomic level here provides an overview of community composition without being too specific, so patterns may be more visible during interpretation.

Why was your faceting choice useful?

Faceting by population allows us to target/interpret differences in community composition based off population status.

Upload screenshot: taxonomic bar plot

Choose File taxbarchart



Which taxa appear to be most abundant?

Proteobacteria!

Did taxonomic composition differ among groups?

Abundance varied across samples and population status.

Part 12 — Overall interpretation

In 3–5 sentences, summarize what your DADA2 results, alpha diversity, beta diversity, and bar plot together suggest about this dataset.

Both Alpha Diversity and Bray-Curtis NMDS plots reveal that samples sharing population status roughly share similar microbiomes. The taxonomic bar plot shows that the microbiomes of different samples vary on the basis of population status, as the data reflects that residents generally possess greater variety and abundance of microbiota by phylum. Alpha diversity also revealed that the highest species richness and evenness tended to correspond with samples of migratory status.

Export (PDF)

1. Click **Prepare submission preview** (generates a clean summary).
2. Click **Download as PDF (Print)** and save.

Submission preview

Dataset
Project/sample set: Bird dataset
Sequencing type: paired-end Illumina
Grouping variable: population

Quality profiles
Purpose: The quality profile tool aids in visualizing and understanding the distribution of respective quality scores.
Reverse vs forward quality: Reverse reads are typically lower quality than forward reads because by the time the first read has been sequenced and the second can start, the signal has degraded and accuracy declines.
Forward quality drop: 210
Reverse quality drop: 170
Forward truncation: 220
Reverse truncation: 200
Why these values: I chose truncation values a margin away from those values whereby quality began to drop in order to keep as much overlap as possible for merging. Willing to tolerate maybe a quality score above 25.

filterAndTrim
Purpose: Trimming from the ends and filtering enhances the accuracy of downstream processing by helping to eliminate sequencing errors.
Parameters used:

```
trimming parameters “truncate read length” set to 220; yes to separate filters and
trimmers for reverse reads; Trim start of each read set to 10; Truncate read lengt
h set to 200;
```

If trimmed too much: Poor merging and loss of usable data can result from overtrimming, reducing coverage/depth of sequence(s).
If not trimmed enough: Low quality sequences can increase error rates.

learnErrors
Purpose: The learnErrors step is important because it determines how and how often errors occur.
Model fit: The fitted model mostly follows the observed error pattern.
Why important: Learning error rates is important to perform before denoising because it allows dada2 to distinguish between true sequence variation and sequencing errors.

dada + mergePairs
Denoising purpose: When provided filtererd reads and error reads, it uses the learned error model to correct errors and reconstruct true sequences (amplicon sequence variants).
ASVs vs OTUs: OTU clustering groups of sequences into taxa based on similarity threshold without modeling errors, whereas ASV Inference corrects sequencing errors at the nucleotide level. ASV provides more precision in VL's opinion.
mergePairs purpose: Forward and reverse reads are being merged to assess confidence of sequence accuracy.
Need for overlap: Overlap between forward and reverse reads confirms that they match and indicates accuracy.
One reason reads fail to merge: Reads may fail to merge if there isn't enough overlap.

Sequence table + sequence length distribution + chimera removal + taxonomy
Sequence table purpose: The ASV table represents diversity with precise comparison of sampled sequences. there are about 1600 ASVs.
Rows/columns/cells: Columns are ASVs, samples, and frequency of ASV appearance in each sample.
Sequence length distribution: Narrow distribution centered mostly around a single peak suggests that the target region was amplified fairly consistently between samples.
Chimera removal purpose: Chimeras are sequences artificially generated during PCR and because they don't actually exist in the original sample the presence of chimeras compromises sequence accuracy and downstream processing.
Why chimeras are a problem: The presence of chimeras can lead to incorrect taxonomic assignment.
Taxonomy purpose: Taxonomy assignment to ASV visually reflects the species and community composition from samples, as well as their roles.
Taxonomy limits: Some ASVs may not be assigned all the way to species due to sequence resolution, reference database limitations, or primarily sequenced regions being too short.

Alpha diversity

Definition: Alpha diversity measures and describes how diverse a particular community is.
Observed: Observed species richness simply measures the total number of species.
Shannon: Shannon (alpha) diversity accounts for richness and evenness by comparing species abundance.
Simpson: Simpson (alpha) diversity accounts for richness and evenness with less sensitivity to rare species, essentially indicating the probability of picking the same species twice.
Differences by population: Generally, residents had a higher alpha diversity than migratory birds.
Differences by sex: Alpha diversity was lower for female birds than male, on average, with there being numerous exceptions.
Differences by flock: If there is any pattern to alpha diversity values here, it shows that flocks very roughly group together.

Beta diversity (NMDS Bray-Curtis)

Definition: Beta diversity is a comparison of one community's composition to that of another.
Bray-Curtis: Individuals (represented by datapoints) with similar microbiomes (respective datapoints will be closer) or dissimilar (respective datapoints will be farther)
NMDS: The NMDS plot shows approximate similarity by sample source in this case, and more generally represents dissimilarity by datapoint proximity.
Patterns by population: The plot shows that there is grouping by beta diversity values around population status.
Patterns by sex: Grouping is less obvious here, but still visible on the basis of sex.
Patterns by flock: Flocks roughly group together around their beta diversity values, with more obvious grouping noticeable from migratory flocks.

Taxonomic bar plot

Taxonomic level: phylum
Faceted by: faceted rows by population
Why this taxonomic level? Using phylum as the taxonomic level here provides an overview of community composition without being too specific, so patterns may be more visible during interpretation.
Why this faceting choice? Faceting by population allows us to target/interpret differences in community composition based off population status.
Most abundant taxa: Proteobacteria!
Did composition differ among groups? Abundance varied across samples and population status.

Overall interpretation

Both Alpha Diversity and Bray-Curtis NMDS plots reveal that samples sharing population status roughly share similar microbiomes. The taxonomic bar plot shows that the microbiomes of different samples vary on the basis of population status, as the data reflects that residents generally possess greater variety and abundance of microbiota by phylum. Alpha diversity also revealed that the highest species richness and evenness tended to correspond with samples of migratory status.